

BONDING – METALLIC BONDING

KEY VOCABULARY

Metallic bond: the strong attraction between positive metal ions and delocalised electrons.

Delocalised: electrons that are free to move through the whole structure.

Lattice: a regular, repeating arrangement of ions.

Malleable: can be hammered or bent into shape without breaking.

COMMON MISCONCEPTIONS

~~✗ Metals are made of molecules.~~

✓ Metals are a giant lattice of ions.

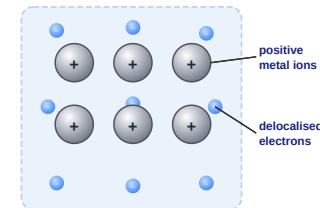
~~✗ The electrons belong to one atom.~~

✓ Electrons are delocalised (shared by all).

~~✗ Metals conduct because ions move.~~

✓ The delocalised electrons carry the charge.

THE METALLIC LATTICE



Positive **metal ions** sit in a 'sea' of **delocalised electrons**.

1

DEFINE THE BOND

Describe what is meant by **metallic bonding**.

Use the words 'positive metal ions' and 'delocalised electrons'.

2

ELECTRICAL CONDUCTIVITY

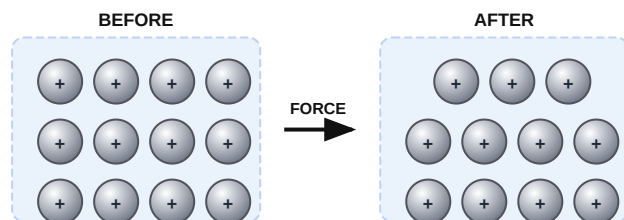
Explain why metals are good conductors of electricity.

Think about which particles are free to move.

3

MALLEABILITY

The diagram shows the lattice before and after a force is applied. **Explain** why the layers of ions can slide.



4

MELTING POINT

Complete the sentence using the word bank.

Metals have _____ melting points because there is a _____ electrostatic _____ between the ions and the delocalised electrons.

high

strong

attraction

5

LABEL THE PARTICLES

In the lattice diagram, what do the **large '+' circles** and the **small '-' circles** represent?

6

WHY SO STRONG?

Suggest why a metal with ions of charge 2+ has a higher melting point than one with 1+ ions.
